

In-Class Problem Set #4: September 2, 2026

1. Bob draws cards (without replacement) from a deck of 52 cards, until he gets an Ace, and then he stops. Suppose that the first Ace appears on the 4th draw. What is the conditional probability that:

- 1a. The first three draws have 0 Kings?
- 1b. The first three draws have 1 King?
- 1c. The first three draws have 2 Kings?
- 1d. The first three draws have 3 Kings?

2. Roll a green 6-sided die and a red 6-sided die one time. Suppose that the sum on the pair of dice is between 4 and 8 (inclusive). What is the conditional probability that:

- 2a. The sum equals 4?
- 2b. The sum equals 5?
- 2c. The sum equals 6?
- 2d. The sum equals 7?
- 2e. The sum equals 8?

3. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six possible colors, namely, red, orange, yellow, green, blue, purple. If we pick such a bear family randomly, the sample space contains all ordered 3-tuples whose elements are r, o, y, g, b, p . Assume that all such outcomes are equally likely. Suppose that a randomly selected bear family has at least one green bear. What is the conditional probability that:

- 3a. The bear family has exactly 1 green bear?
- 3b. The bear family has exactly 2 green bears?
- 3c. The bear family has exactly 3 green bears?

4. In every round of a game, Alice and Bob roll a die simultaneously. Alice's die is 6-sided. Bob's die is 4-sided. Consider a round in which Bob's result is strictly smaller than Alice's result. On that round, what is the conditional probability that:

- 4a. Bob's die shows the value 1?
- 4b. Bob's die shows the value 2?
- 4c. Bob's die shows the value 3?
- 4d. Bob's die shows the value 4?

5. In every round of a game, Alice and Bob (again!) roll a die simultaneously. Alice's die is (again!) 6-sided. Bob's die is (again!) 4-sided. Consider a round in which Alice's result is strictly smaller than Bob's result. On that round, what is the conditional probability that:

- 5a. Alice's die shows the value 1?
- 5b. Alice's die shows the value 2?
- 5c. Alice's die shows the value 3?